Nth root of a complex number

Basic skills

Q1. $z = 2(1+i) = 2+2i$

a) $|z| = \sqrt{2^2+2^2} = \sqrt{8} = 2\sqrt{2}$

$\text{Arg}\, z = \tan^{-1}\left(\frac{2}{2}\right) = \frac{\pi}{4}$ so $z = 2\sqrt{2}e^{\frac{\pi}{4}i}$

b) $z^2 = \left(2\sqrt{2}e^{\frac{\pi}{4}i}\right)^2$

$= 8e^{\frac{\pi}{2}i}$

c) $z^6 = (z^2)^3 = \left(8e^{\frac{\pi}{2}i}\right)^3$

$= 512e^{\frac{3\pi}{2}i} \xrightarrow{-2\pi} = 512e^{-\frac{\pi}{2}i}$

d) convert back

$z^6 = 512\left(\cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)\right)$

$= 512(0 - i) = -512i$

$a = 0 \qquad b = -512$

Q2. $z = 2e^{\frac{\pi}{3}i}$ $w = 2e^{-\frac{2\pi}{3}i}$

a) $z^2 = \left(2e^{\frac{\pi}{3}i}\right)^2 = 4e^{\frac{2\pi}{3}i}$ same

b) $w^2 = \left(2e^{-\frac{2\pi}{3}i}\right)^2 = 4e^{-\frac{4\pi}{3}i} = 4e^{\frac{2\pi}{3}i}$

c) so $v^2 = 4e^{\frac{2}{3}\pi i}$

has z and w as solutions

Competency skills

Q3. a) $z^4 = 81e^{\frac{\pi}{8}i}$

$w = 81e^{\frac{\pi}{8}i}$ $w_k = 81e^{\left(\frac{\pi}{8}+2\pi k\right)i}$ $k=0,1,2,3$

Then $z_k = \sqrt[4]{81}\, e^{\frac{\left(\frac{\pi}{8}+2\pi k\right)i}{4}}$

so $z_k = 3e^{\left(\frac{\pi}{32}+\frac{\pi}{2}k\right)i}$ $k=0,1,2,3$

$z_0 = 3e^{\frac{\pi}{32}i}$ $z_1 = 3e^{\frac{17\pi}{32}i}$

$z_2 = 3e^{\frac{33\pi}{32}i} \xrightarrow{-2\pi} = 3e^{-\frac{31\pi}{32}i}$ $z_3 = 3e^{\frac{49\pi}{32}i} \overset{-2\pi}{=} 3e^{-\frac{14}{32}i}$

Could use $k=-1,-2$ instead of $2,3$

b) $z^3 = 20e^{i}$

$w = 20e^{1i}$ $\quad w_k = 20e^{(1+2\pi k)i}$ $\quad k=0,1,2$

then

$z_k = \sqrt[3]{20}\, e^{\left(\frac{1}{3}+\frac{2\pi}{3}k\right)i}$ $\quad k=0,1,2$

$z_0 = \sqrt[3]{20}\, e^{\frac{1}{3}i}$ $\quad z_1 = \sqrt[3]{20}\, e^{\left(\frac{1}{3}+\frac{2\pi}{3}\right)i}$

$z_2 = \sqrt[3]{20}\, e^{\left(\frac{1}{3}+\frac{4\pi}{3}\right)i} \xrightarrow{-2\pi} = \sqrt[3]{20}\, e^{\left(\frac{1}{3}-\frac{2\pi}{3}\right)i}$

c) $z^5 = 7-8i$

$w = 7-8i$ ← need exponential form

$|w| = \sqrt{7^2+8^2} = \sqrt{113}$

$\operatorname{Arg} w = -0.719$ so $w = \sqrt{113}\, e^{-0.719i}$

$w_k = \sqrt{113}\, e^{(2\pi k - 0.719)i}$ $\quad k=0,1,..,4$

so $z_k = \sqrt[10]{113}\, e^{\left(\frac{2\pi}{5}k - 0.144\right)i}$ $\quad k=0,1,..,4$

$\sqrt[5]{\sqrt[2]{\ }} = \sqrt[10]{\ }$

$z_0 = \sqrt[10]{113}\, e^{-0.144i}$

$z_1 = \sqrt[10]{113}\, e^{\left(\frac{2\pi}{5} - 0.144\right)i}$

$z_2 = \sqrt[10]{113}\, e^{\left(\frac{4\pi}{5} - 0.144\right)i}$

-2π

$z_3 = \sqrt[10]{113}\, e^{\left(\frac{6\pi}{5} - 0.144\right)i} = \sqrt[10]{113}\, e^{-\left(\frac{4\pi}{5} + 0.144\right)i}$

-2π

$z_4 = \sqrt[10]{113}\, e^{\left(\frac{8\pi}{5} - 0.144\right)i} = \sqrt[10]{113}\, e^{-\left(\frac{2\pi}{5} + 0.144\right)i}$

Q4. For $z = \sqrt[3]{i-1}$

solve $z^3 = i - 1$

$w = i - 1 \qquad |w| = \sqrt{2} \qquad \operatorname{Arg} w = \frac{3\pi}{4}$

careful as real<0

so $w = \sqrt{2}\, e^{\frac{3\pi}{4}i}$

$w_k = \sqrt{2}\, e^{\left(\frac{3\pi}{4} + 2\pi k\right)i}$

$k = 0, 1, 2$

2 out range

so $z_k = \sqrt[6]{2}\, e^{\left(\frac{\pi}{4} + \frac{2\pi}{3}k\right)i}$

$k = 0, 1, -1$

$z_0 = \sqrt[6]{2}\, e^{\frac{\pi}{4} i}$

$z_1 = \sqrt[6]{2}\, e^{\frac{11}{12}\pi i}$

z_{-1} (or z_2) $= \sqrt[6]{2}\, e^{-\frac{5}{12}\pi i}$

Q5. a) $z^5 = 1$ ← 5^{th} roots of unity

$w = 1 \quad |w| = 1 \quad \operatorname{Arg} w = 0$

$w = e^{0i}$

$w_k = e^{2\pi k i} \quad k = 0, 1, \dots, 4$

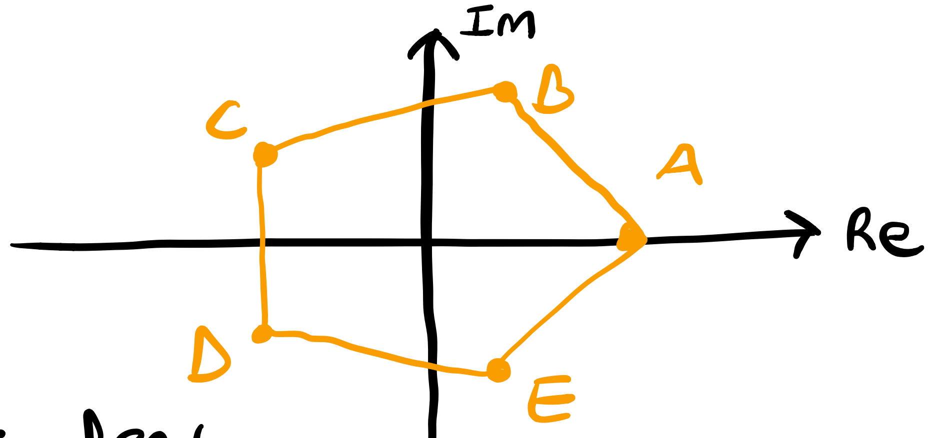
$z_k = e^{\frac{2\pi}{5} k i} \quad k = 0, 1, \dots, 4$

$z_0 = e^{0i} = 1$

$z_1 = e^{\frac{2\pi}{5} i} \qquad z_2 = e^{\frac{4\pi}{5} i}$

$z_3 = e^{\frac{6\pi}{5} i} = e^{-\frac{4\pi}{5} i} \qquad z_4 = e^{\frac{8\pi}{5} i} = e^{-\frac{2\pi}{5} i}$

b)



c) regular Pentagon

Advanced skills

EXAM PAPERS PRACTICE

Q6. $(2z+3)^3 = 3+4i$

let $v = 2z+3$ $\left(\text{so } z = \frac{v-3}{2}\right)$

$w = 3+4i$

then solve $v^3 = w$

$|w| = 5$ $\quad \text{Arg } w = \tan^{-1}\left(\frac{4}{3}\right) = 0.927$

$$w = 5e^{0.927i}$$

$$w_k = 5e^{(0.927+2\pi k)i}$$

$k = 0, 1, 2$

so $$v_k = \sqrt[3]{5}\, e^{\left(0.309+\frac{2\pi}{3}k\right)i}$$

$k = 0, 1, 2$

can use $k = -1$

$$v_0 = \sqrt[3]{5}\, e^{0.309i}$$

$$v_1 = \sqrt[3]{5}\, e^{\left(0.309+\frac{2\pi}{3}\right)i}$$

$$v_2 = \sqrt[3]{5}\, e^{\left(0.309-\frac{2\pi}{3}\right)i}$$

now to find z using $v = 2z+3$

so $z = \frac{v-3}{2}$ but easier in normal form

so convert by mod-arg

$v_0 = 1.63 + 1.15i$

$v_1 = -1.26 + 0.52i$

$v_2 = -0.36 - 1.67i$

Then $z_k = \frac{v_k - 3}{2}$ ← only subtract from real

$z_0 = -0.685 + 0.575i$

$z_1 = -2.13 + 0.26i$

$z_2 = -1.68 - 0.835i$

Q7. a) To show $e^{\frac{2\pi}{5}i}$ is a fifth root of unity $\left(e^{\frac{2\pi}{5}i}\right)^5 = e^{2\pi i} = 1$

b) 1 is clearly a fifth root of unity

For others

$w^2 = e^{\frac{4\pi}{5}i}$ then $\left(e^{\frac{4\pi}{5}i}\right)^5 = e^{4\pi i} = 1$

$w^3 = e^{\frac{6\pi}{5}i}$ then $\left(e^{\frac{6\pi}{5}i}\right)^5 = e^{6\pi i} = 1$

$w^4 = e^{\frac{8\pi}{5}i}$ then $\left(e^{\frac{8\pi}{5}i}\right)^5 = e^{8\pi i} = \underline{1}$

c)

$1 + w + w^2 + w^3 + w^4$

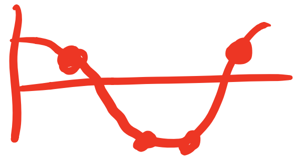
$= 1 + e^{\frac{2\pi}{5}i} + e^{\frac{4\pi}{5}i} + e^{\frac{6\pi}{5}i} + e^{\frac{8\pi}{5}i}$

$= 1 + \cos\left(\frac{2\pi}{5}\right) + i\sin\left(\frac{2\pi}{5}\right)$

$+ \cos\left(\frac{4\pi}{5}\right) + i\sin\left(\frac{4\pi}{5}\right)$

$+ \cos\left(\frac{6\pi}{5}\right) + i\sin\left(\frac{6\pi}{5}\right)$

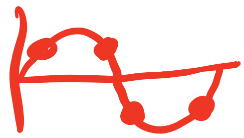
$+ \cos\left(\frac{8\pi}{5}\right) + i\sin\left(\frac{8\pi}{5}\right)$

now by symmetry 

$$\cos\left(\frac{2\pi}{5}\right) = \cos\left(\frac{8\pi}{5}\right) = 0.309\ldots$$

$$\cos\left(\frac{4\pi}{5}\right) = \cos\left(\frac{5\pi}{5}\right) = -0.809$$

and



$$\sin\left(\frac{2\pi}{5}\right) = -\sin\left(\frac{6\pi}{5}\right)$$

$$\sin\left(\frac{4\pi}{5}\right) = -\sin\left(\frac{8\pi}{5}\right)$$

so immanigary disappears to give

$$1 + w + w^2 + w^3 + w^4 =$$

$$1 + 0.309 + 0.309 - 0.809 - 0.809 = 0$$

d) $w^2 + w^5 + w^8 + w^{11} + w^{-1}$

As 5^{th} roots of unity can increase power by ± 5

so $w^2 + w^0 + w^3 + w^1 + w^4$

$\ \ 5-5 \quad 8-5 \quad 11-10 \quad -1+5$

$= 0$

Q8. a) $z^3 = 2+2i$

$w = 2+2i \qquad |w| = 2\sqrt{2} \qquad \operatorname{Arg} w = \frac{\pi}{4}$

$w = 2\sqrt{2}e^{\frac{\pi}{4}i}$ so

$$w_k = 2\sqrt{2}\,e^{\left(\frac{\pi}{4}+2\pi k\right)i} \qquad k=0,1,2$$

$$z_k = \sqrt[3]{2\sqrt{2}}\,e^{\left(\frac{\pi}{12}+\frac{2\pi}{3}k\right)i} \qquad k=0,1,2$$

$$= \sqrt[6]{8}\,e^{\left(\frac{\pi}{12}+\frac{2\pi}{3}k\right)i} \qquad k=0,1,2$$

so $z_0 = \sqrt[6]{8}\,e^{\frac{\pi}{12}i}$

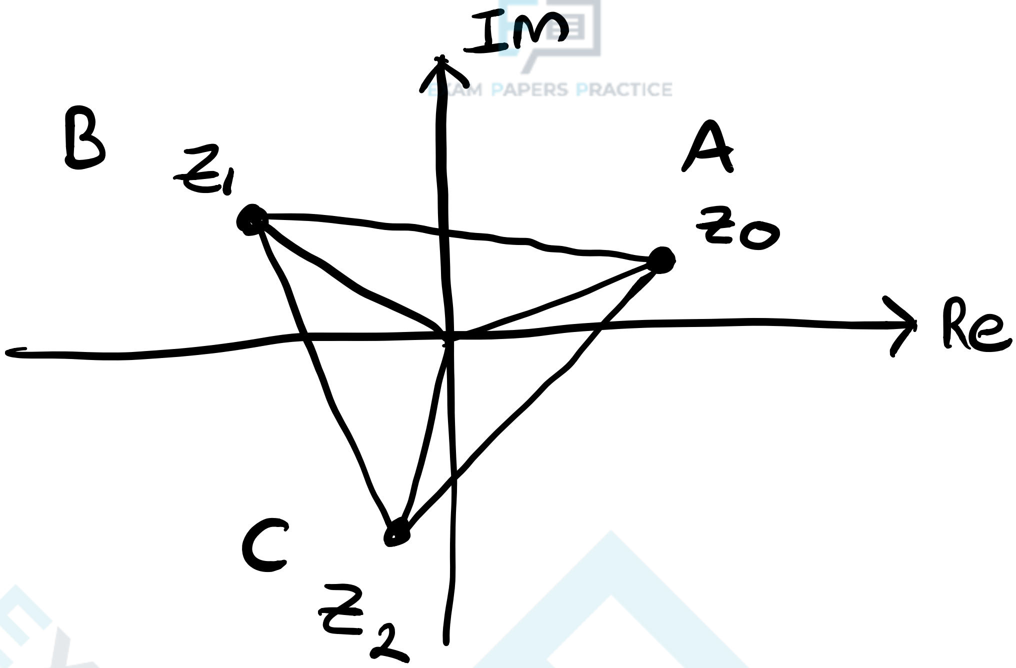
$z_1 = \sqrt[6]{8}\,e^{\frac{3\pi}{4}i}$

$z_2 = \sqrt[6]{8}\,e^{-\frac{7\pi}{12}i}$

or $k=-1$ gives same

so $|z_k| = \sqrt[6]{8}$

$$\operatorname{Arg} z_k = \begin{cases} \frac{\pi}{12} & k=0 \\ \frac{3\pi}{4} & k=1 \\ -\frac{7\pi}{12} & k=2 \end{cases}$$



c) As the triangle is equilateral (regular 3-gon) all we need is a distance between two points

AB is $|z_1 - z_0|$

$z_0 = \sqrt[6]{8}\left(\cos\left(\frac{\pi}{12}\right) + \sin\left(\frac{\pi}{12}\right)i\right)$

$= 1.366 + 0.366i$

$z_1 = \sqrt[6]{8}\left(\cos\left(\frac{3\pi}{4}\right) + \sin\left(\frac{3\pi}{4}\right)i\right)$

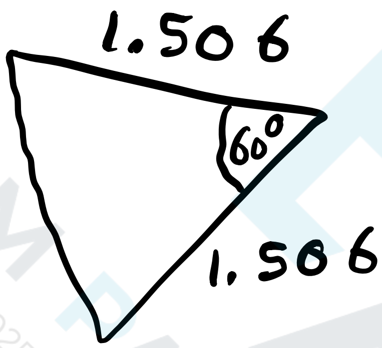
$= 0 + i$

so $|z_1 - z_0| = |0.634i - 1.366|$

$$= \sqrt{0.634^2 + 1.366^2}$$

$$= 1.506$$

so



so $\frac{1}{2}ab\sin C$

$$\frac{1}{2} \times 1.506^2 \times \sin 60°$$

$$= 0.982$$

The enscribed circle is of radius $\sqrt[6]{8}$ so area is

$$\frac{1}{2}\pi\left(\sqrt[6]{8}\right)^2 = \pi \approx 3.14$$

$0.982 < 3.14$ so yes